



國立陽明交通大學

NATIONAL YANG MING CHIAO TUNG UNIVERSITY

前瞻積體電路設計實驗室

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## ADFP Cell-based Training Course

ICLAB 特別課程實作：

16奈米 FinFET 技術下的異步 FIFO 設計

Report

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## Overview

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In this lab, students will follow a comprehensive IC design flow to gain hands-on experience with advanced FinFET technology. Each step in the process builds on the previous one, ensuring a clear relationship that enhances both understanding and skill development.

1. **RTL Design (01\_RTL):** Students begin by designing the RTL code for a simple FIFO structure, learning how to describe hardware behavior using Verilog.
2. **Logic Synthesis (02\_SYN):** The RTL design is synthesized into gate-level netlists using Design Compiler, helping students understand how high-level designs are translated into actual hardware.
3. **Gate-Level Simulation (03\_GATE):** Students simulate the synthesized design to verify its functional correctness, ensuring that the gate-level implementation matches the intended RTL behavior.
4. **Hard Macro SRAM (04\_MEM):** The lab introduces the integration of hard macro SRAMs into the design, teaching students how to use pre-built memory blocks to improve performance and area efficiency.
5. **Physical Design (05\_APR):** Through Auto Placement and Routing (APR), students learn how to map the netlist into a physical layout, addressing challenges such as cell placement, routing, and the integration of boundary cells and tap cells.
6. **Post Layout Simulation (06\_POST):** Students simulate the post-layout design to verify its timing and functional correctness, learning the importance of layout impact on performance.
7. **Power Analysis (07\_PX):** Using PrimeTime, students perform dynamic and leakage power analysis, comparing results to understand the effect of optimizations like clock gating.
8. **CDC Formal Verification (08\_JG):** The lab covers Clock Domain Crossing (CDC) verification using JasperGold, teaching students how to ensure proper data transfer between asynchronous domains.
9. **Sign-off Checks (09\_RVE):** The flow concludes with Design Rule Check (DRC) and Layout Versus Schematic (LVS) verification, emphasizing the critical final steps for ensuring design manufacturability and correctness.

By completing this flow, students will gain a practical understanding of each phase in the IC design process, from RTL to sign-off, and develop the skills necessary to tackle real-world challenges in advanced FinFET technology.

**Design:** Pseudo Random Number Generator (PRNG)**Data Preparation**

- Extract LAB data from TA's directory:  

```
% tar -xvf ~ba25iclabTA01/Lab_adfp_2025F.tar
```
- The extracted LAB directory contains:
  - a. UMC018/
  - b. N16ADFP/
  - c. N16ADFP\_cg/
  - d. N16ADFP\_cg\_chip/
  - e. src/

**Design Description**

A Pseudo-Random Number Generator (PRNG) is a deterministic algorithm capable of producing a sequence of numbers that approximates the properties of random numbers. Unlike True Random Number Generators (TRNG), which rely on physical entropy, PRNGs expand a small initial value, known as the Seed, into a much longer sequence.

The generation of random sequences in this design is implemented using the Xorshift family of Pseudo-Random Number Generators (PRNGs). While numerous algorithms exist for random number generation, Xorshift variants are specifically selected due to their renowned simplicity and execution speed, particularly on modern hardware architectures.

|       |       |       |       |
|-------|-------|-------|-------|
| 95506 | 26237 | 84098 | 35156 |
| 14294 | 55172 | 20805 | 2732  |
| 46292 | 73243 | 49033 | 74764 |
| 62320 | 66361 | 15910 | 55492 |
| 74695 | 48060 | 52440 | 45819 |
| 73448 | 78340 | 70253 | 58152 |

Figure 1. Random Number example

In this lab, you are asked to build a design to perform Pseudo Random Number Generator (PRNG) to calculate 256 random numbers by the given seed for each pattern.

- Block diagram

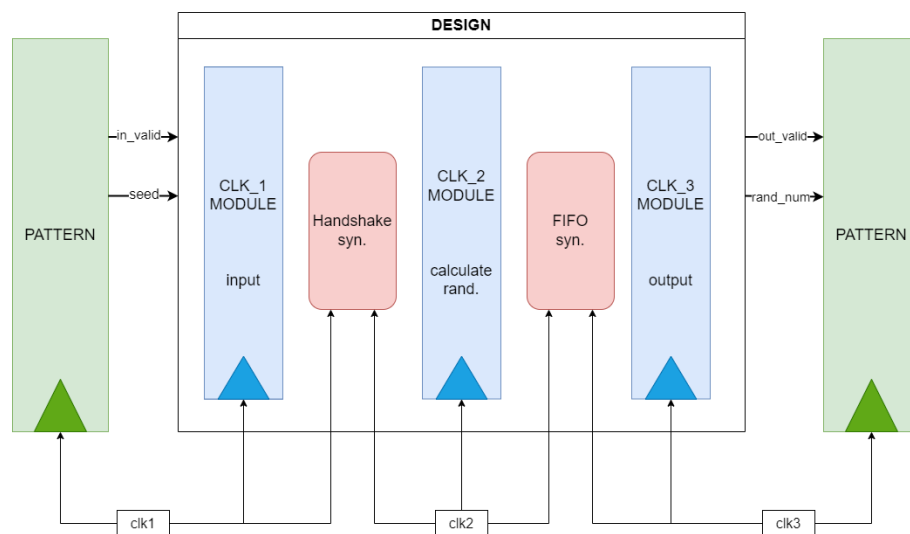


Figure 2. The block diagram of this design in this lab

1. The seed is given in clk1 domain. (Serial in: 1-bit for 32 cycles)
2. Use Handshake synchronizer to transfer the seed into clk2 domain.
3. Calculate the random number in clk2 domain.
4. Store the result into FIFO synchronizer and get the random number in clk3 domain.
5. Output the random number in the clk3 domain (Serial out: 1-bit for 32 cycles)

- Algorithm of PRNG: Xorshift algorithm

1. Initialization: start with a non-zero 32-bit seed  $X_0$  (1-bit serial in)
2. Generation: for each desired random number
  - a.  $X_{n+1} = X_n$
  - b.  $X_{n+1} = X_{n+1} \oplus (X_{n+1} \ll a)$  (left shift and XOR)
  - c.  $X_{n+1} = X_{n+1} \oplus (X_{n+1} \gg b)$  (right shift and XOR)
  - d.  $X_{n+1} = X_{n+1} \oplus (X_{n+1} \ll c)$  (left shift and XOR)
  - e. Return  $X_{n+1}$

where a, b, c are constants.

In this lab, we use 32-bit version, that is, a = 13, b = 17 and c = 5

- CDC problem
  1. The Handshake synchronizer is used to cross clk1 to clk2.
  2. The FIFO synchronizer is used to cross clk2 to clk3.
- Cycle time

|         | CYCLE 1 | CYCLE 2 | CYCLE 3 |
|---------|---------|---------|---------|
| UMC018  | 14.1 ns | 3.9 ns  | 20.7 ns |
| N16ADFP | 1.41 ns | 0.49 ns | 2.13 ns |

## I/O Description

| Input     | Bits | Definition and Description                                                                                                                                                                  |
|-----------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| clk1      | 1    | Positive edge trigger clock by clock 1<br>clock period: UMC018=14.1, N16ADFP=1.41 (ns)                                                                                                      |
| clk2      | 1    | Positive edge trigger clock by clock 2<br>clock period: UMC018=3.9, N16ADFP=0.49 (ns)                                                                                                       |
| clk3      | 1    | Positive edge trigger clock by clock 3<br>clock period: UMC018=20.7, N16ADFP=2.07 (ns)                                                                                                      |
| rst_n     | 1    | Asynchronous reset active low reset                                                                                                                                                         |
| in_valid  | 1    | seed are valid when in_valid is high This signal is trigger by clk1 for 32 cycles to serial in for a 32-bit value (MSB to LSB)                                                              |
| seed      | 1    | The seed for Xorshift algorithm. A positive number $X_0$                                                                                                                                    |
| Output    | Bits | Definition and Description                                                                                                                                                                  |
| out_valid | 1    | should be set to low after reset and not be raised when invalid is high. Should set to high when your rand_num is ready. Should be pulled up for 256 cycles, not required to be continuous. |
| rand_num  | 1    | The generated number $X_1, X_2, \dots, X_{256}$ should be output for 256 cycles, not required to be continuous. Each 32-bit X will be serial out for 32 cycles (MSB to LSB)                 |

## Front-End Design Steps

### A. 01\_RTL

1. Finish “**DESIGN\_module.v**” in src/01\_RTL

- |      |                                                                  |
|------|------------------------------------------------------------------|
| i.   | $X_{n+1} = X_n$                                                  |
| ii.  | $X_{n+1} = X_{n+1} \oplus (X_{n+1} \ll a)$ (left shift and XOR)  |
| iii. | $X_{n+1} = X_{n+1} \oplus (X_{n+1} \gg b)$ (right shift and XOR) |
| iv.  | $X_{n+1} = X_{n+1} \oplus (X_{n+1} \ll c)$ (left shift and XOR)  |
| v.   | Return $X_{n+1}$                                                 |



```
//=====//
//      Finish this part      //
//=====//

// shift temp
always @(*) begin
  case (state)
    1: shift_temp = ;
    2: shift_temp = ;
    3: shift_temp = ;
    default: shift_temp = 0;
  endcase
end
assign share_temp_next = ;
//=====//
```

2. Open “**FIFO\_syn.v**” in src/01\_RTL/synchronizer
  - a. For design, what kind of SRAM we use? ☐ Dual-Port / ☒ Two-Port
  - b. Can we read & write to the same address at the same time? ☐ Yes / ☒ No
3. Open “**PRNG\_TOP.v**” in src/01\_RTL and “**PATTERN.v**” in src/00\_TESTBED
  - a. If we use **synchronous reset**, we have to pass the **setup/hold** time check.
  - b. If we use **asynchronous reset**, we should pass the **recovery/removal** time check
  - c. In this lab, ☒ Asynchronous / ☐ Synchronous reset technique is adopted.
  - d. In real case, do we usually force  $\text{clk} = 1'b0$  when  $\text{rst\_n} = 0$ ? ☐ Yes / ☒ No

Therefore, reset synchronizers are used to solve the metastability issue between  $\text{clk}$  &  $\text{rst}$

4. Run “**./01\_run\_vcs\_rtl**” to do RTL-level simulation and pass all pattern

### B. 02\_SYN

1. Standard cell library comparison for different manufacturing processes
  - a. For corner case, **S, T and F means slow, typical and fast**, respectively, which is used to represent the speed of MOS. For example, FF, SS, TT, FS and SF corners are used for simulation. The first one is for NMOS, the second one is for PMOS.
  - b. For UMC018 process, the library files are placed in the following path:

**/RAID2/PROCESS/U18/UMC018\_CBDK/CIC/SynopsysDC/lib/**

Check the three files listed below to observe different PVT situation

- **fsa0m\_a\_generic\_core\_ff1p98vm40c.lib** (FF corner, 1.98V, -40°C)
- **fsa0m\_a\_generic\_core\_tt1p8v25c.lib** (TT corner, 1.80V, 25°C)

- **fsa0m\_a\_generic\_core\_ss1p62v125c.lib** (SS corner, 1.62V, 125°C)

c. You can see the definition of index & variable of the top of each .lib

```
variable_1 : input_net_transition;
variable_2 : total_output_net_capacitance;
index_1 ("0.0, 1.0, 2.0, 3.0, 4.0, 5.0");
index_2 ("0.0, 1.0, 2.0, 3.0, 4.0, 5.0");
```

d. Find the NAND2 cell specified below and observe power & timing information

- NAND2 cell: **ND2**
- pin(O), related pin: "I1", (index1, index2) = (0,0)
- Lookup table for power information (internal power)
- Lookup table for timing information (rising time)
- Finish the table below and answer the question

| Corner | Voltage | Temperature | Power    | Delay    |
|--------|---------|-------------|----------|----------|
| FF     | 1.98V   | 40°C        | 0.010103 | 0.021058 |
| TT     | 1.80V   | 25°C        | 0.008827 | 0.030047 |
| SS     | 1.62V   | 125°C       | 0.007664 | 0.047627 |

**Question:** Which cell should be used for worse case analysis?

☐ ff / ☐ tt / ☒ ss, why: 電壓低、溫度高、delay高，所以是最壞情況

e. For N16ADFP process, the library files are placed in the following path:

**/RAID2/PROCESS/ADFP/Executable\_Package/Collaterals/IP/stdcell/N16ADFP\_StdCell/CCS/**

Check the five files listed below to observe different PVT situation

- **N16ADFP\_StdCellff0p88v125c\_ccs.lib** (FF corner, 0.88V, 125°C)
- **N16ADFP\_StdCellff0p88vm40c\_ccs.lib** (FF corner, 0.88V, -40°C)
- **N16ADFP\_StdCelltt0p8v25c\_ccs.lib** (TT corner, 0.80V, 25°C)
- **N16ADFP\_StdCellss0p72v125c\_ccs.lib** (SS corner, 0.72V, 125°C)
- **N16ADFP\_StdCellss0p72vm40c\_ccs.lib** (SS corner, 0.72V, -40°C)

In ADFP process, standard threshold voltage (SVT) cells and low threshold voltage (LVT) cells are provided for tool to synthesize and optimize circuit.

f. Find the NAND2 cell specified below and observe power & timing information

- SVT NAND2: **ND2D12BWP16P90**



- LVT NAND2: ND2D12BWP16P90LVT
- Power information (leakage power)
- Timing information (rising time)
- Finish the table below and answer the question

| Corner   | Voltage | Temperature | Power    | Delay      |
|----------|---------|-------------|----------|------------|
| FF (lvt) | 0.88V   | 40°C        | 3.63197  | 0.00211245 |
| FF (lvt) | 0.88V   | 125°C       | 782.776  | 0.00219008 |
| FF (svt) | 0.88V   | 40°C        | 0.179304 | 0.00249980 |
| FF (svt) | 0.88V   | 125°C       | 107.562  | 0.00250904 |
| TT (lvt) | 0.80V   | 25°C        | 23.216   | 0.00230    |
| TT (svt) | 0.80V   | 25°C        | 1.368    | 0.00282    |
| SS (lvt) | 0.72V   | 40°C        | 0.556901 | 0.00256213 |
| SS (lvt) | 0.72V   | 125°C       | 232.433  | 0.00256762 |
| SS (svt) | 0.72V   | 40°C        | 0.02017  | 0.00348809 |
| SS (svt) | 0.72V   | 125°C       | 23.1403  | 0.00328336 |

**Question:** Which cell have higher leakage power?

■ lvt / □ svt, why: V<sub>th</sub>小，漏電流為指數上升

2. Run “./01\_run\_dc\_shell” to synthesize the circuit (in UMC018 & N16ADFP)

- a. Area comparison

|      | UMC018        | N16ADFP     |
|------|---------------|-------------|
| Area | 224589.013951 | 3687.690846 |

The design area in UMC018 is 60.9 times larger than that in N16ADFP

- b. Paste your synthesis area report here

```

*****
Report : area
Design : PRNG_TOP
Version: X-2025.06
Date   : Thu Dec 25 06:18:59 2025
*****

Library(s) Used:

N16ADFP StdCells0p72vm40c.ccs (File: /RAID2/PROCESS/ADFP/Executable Package/Collaterals/IP
N16ADFP SRAM_ss0p72v0p72vm40c (File: /RAID2/COURSE/a25/iclab/6a25iclab09/Lab_adfp_2025F/N1

Number of ports:      89
Number of nets:      954
Number of cells:      857
Number of combinational cells: 568
Number of sequential cells: 265
Number of macros/black boxes: 1
Number of buf/inv:    96
Number of references: 58

Combinational area:   181.854724
Buf/Inv area:         15.811201
Noncombinational area: 383.782411
Macro/Black Box area: 3282.053711
Net Interconnect area: undefined (Wire load has zero net area)

Total cell area:      3687.690846
Total area:           undefined

```

```

*****
Report : area
Design : PRNG_TOP
Version: X-2025.06
Date   : Thu Dec 25 06:23:32 2025
*****

Library(s) Used:

fsa0m_a_generic_core_ss1p62v125c (File: /RAID2/PROCESS/UI8/UMC018 CBDK/CIC/Synopsys
S_JMA180_128X32X1BM1_MC (File: /RAID2/COURSE/a25/iclab/6a25iclab09/Lab_adfp_2025F/

Number of ports:      89
Number of nets:      995
Number of cells:      893
Number of combinational cells: 606
Number of sequential cells: 265
Number of macros/black boxes: 1
Number of buf/inv:    128
Number of references: 55

Combinational area:   18077.479993
Buf/Inv area:         1899.929588
Noncombinational area: 15827.112083
Macro/Black Box area: 198684.421875
Net Interconnect area: undefined (Wire load has zero net area)

Total cell area:      224589.013951
Total area:           undefined

```

3. Run “./02\_run\_pt” to disable setup & hold time check for synchronizers

## C. 02\_SYN with cg

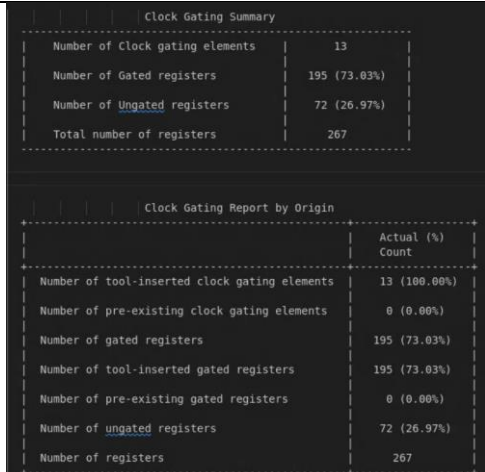
1. Observe commands we set in N16ADFP\_cg/02\_SYN/syn.tcl about clock gating, and as the gating result we get

- a. What is the clock gating cells? CKLNQD1BWP20P90
- b. We gate 73.03 % of registers?
- c. How do you think about this gating method? 可避免glitch的產生，也可以降低dynamic power
- d. There are many gating techniques, refer to the table and match each technique:
- Clock Gating: (1)      Multi-Threshold Gating: (4)
- Power Gating: (2)      Data Gating: (3)

| Gating Technique | Time Overhead    | Power Savings | Complexity | Use Case                                                        |
|------------------|------------------|---------------|------------|-----------------------------------------------------------------|
| (1)              | Low              | Moderate      | Low        | Frequently used for reducing dynamic power in digital circuits. |
| (2)              | Moderate to High | High          | High       | Used in low-power design for shutting down idle blocks.         |
| (3)              | Low              | Moderate      | Moderate   | Efficient in reducing unnecessary data toggling                 |
| (4)              | Moderate         | High          | High       | Common in designs requiring fine-grained power control.         |

2. Run “./01\_run\_dc\_shell” in N16ADFP\_cg directory and observe the result in N16ADFP\_cg/02\_SYN/Report/PRNG\_TOP.cg.

Paste your result in the box below

| Example                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Your Result                     |    |                           |              |                             |              |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |              |                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |    |                           |              |                             |             |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |             |                     |     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|----|---------------------------|--------------|-----------------------------|--------------|---------------------------|-----|--|------------------|-----------------------------------------------|--------------|----------------------------------------------|-----------|---------------------------|--------------|-----------------------------------------|--------------|----------------------------------------|-----------|-----------------------------|--------------|---------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|----|---------------------------|--------------|-----------------------------|-------------|---------------------------|-----|--|------------------|-----------------------------------------------|--------------|----------------------------------------------|-----------|---------------------------|--------------|-----------------------------------------|--------------|----------------------------------------|-----------|-----------------------------|-------------|---------------------|-----|
| <p><b>Clock Gating Summary</b></p> <table> <tr> <td>Number of Clock gating elements</td><td>11</td></tr> <tr> <td>Number of Gated registers</td><td>185 (64.46%)</td></tr> <tr> <td>Number of Ungated registers</td><td>102 (35.54%)</td></tr> <tr> <td>Total number of registers</td><td>287</td></tr> </table> <p><b>Clock Gating Report by Origin</b></p> <table> <tr> <th></th><th>Actual (%) Count</th></tr> <tr> <td>Number of tool-inserted clock gating elements</td><td>11 (100.00%)</td></tr> <tr> <td>Number of pre-existing clock gating elements</td><td>0 (0.00%)</td></tr> <tr> <td>Number of gated registers</td><td>185 (64.46%)</td></tr> <tr> <td>Number of tool-inserted gated registers</td><td>185 (64.46%)</td></tr> <tr> <td>Number of pre-existing gated registers</td><td>0 (0.00%)</td></tr> <tr> <td>Number of ungated registers</td><td>102 (35.54%)</td></tr> <tr> <td>Number of registers</td><td>287</td></tr> </table> | Number of Clock gating elements | 11 | Number of Gated registers | 185 (64.46%) | Number of Ungated registers | 102 (35.54%) | Total number of registers | 287 |  | Actual (%) Count | Number of tool-inserted clock gating elements | 11 (100.00%) | Number of pre-existing clock gating elements | 0 (0.00%) | Number of gated registers | 185 (64.46%) | Number of tool-inserted gated registers | 185 (64.46%) | Number of pre-existing gated registers | 0 (0.00%) | Number of ungated registers | 102 (35.54%) | Number of registers | 287 |  <p><b>Clock Gating Summary</b></p> <table> <tr> <td>Number of Clock gating elements</td><td>13</td></tr> <tr> <td>Number of Gated registers</td><td>195 (73.03%)</td></tr> <tr> <td>Number of Ungated registers</td><td>72 (26.97%)</td></tr> <tr> <td>Total number of registers</td><td>267</td></tr> </table> <p><b>Clock Gating Report by Origin</b></p> <table> <tr> <th></th><th>Actual (%) Count</th></tr> <tr> <td>Number of tool-inserted clock gating elements</td><td>13 (100.00%)</td></tr> <tr> <td>Number of pre-existing clock gating elements</td><td>0 (0.00%)</td></tr> <tr> <td>Number of gated registers</td><td>195 (73.03%)</td></tr> <tr> <td>Number of tool-inserted gated registers</td><td>195 (73.03%)</td></tr> <tr> <td>Number of pre-existing gated registers</td><td>0 (0.00%)</td></tr> <tr> <td>Number of ungated registers</td><td>72 (26.97%)</td></tr> <tr> <td>Number of registers</td><td>267</td></tr> </table> | Number of Clock gating elements | 13 | Number of Gated registers | 195 (73.03%) | Number of Ungated registers | 72 (26.97%) | Total number of registers | 267 |  | Actual (%) Count | Number of tool-inserted clock gating elements | 13 (100.00%) | Number of pre-existing clock gating elements | 0 (0.00%) | Number of gated registers | 195 (73.03%) | Number of tool-inserted gated registers | 195 (73.03%) | Number of pre-existing gated registers | 0 (0.00%) | Number of ungated registers | 72 (26.97%) | Number of registers | 267 |
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| Number of Gated registers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 185 (64.46%)                    |    |                           |              |                             |              |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |              |                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |    |                           |              |                             |             |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |             |                     |     |
| Number of Ungated registers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 102 (35.54%)                    |    |                           |              |                             |              |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |              |                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |    |                           |              |                             |             |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |             |                     |     |
| Total number of registers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 287                             |    |                           |              |                             |              |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |              |                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |    |                           |              |                             |             |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |             |                     |     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Actual (%) Count                |    |                           |              |                             |              |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |              |                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |    |                           |              |                             |             |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |             |                     |     |
| Number of tool-inserted clock gating elements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 11 (100.00%)                    |    |                           |              |                             |              |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |              |                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |    |                           |              |                             |             |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |             |                     |     |
| Number of pre-existing clock gating elements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0 (0.00%)                       |    |                           |              |                             |              |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |              |                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |    |                           |              |                             |             |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |             |                     |     |
| Number of gated registers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 185 (64.46%)                    |    |                           |              |                             |              |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |              |                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |    |                           |              |                             |             |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |             |                     |     |
| Number of tool-inserted gated registers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 185 (64.46%)                    |    |                           |              |                             |              |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |              |                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |    |                           |              |                             |             |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |             |                     |     |
| Number of pre-existing gated registers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0 (0.00%)                       |    |                           |              |                             |              |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |              |                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |    |                           |              |                             |             |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |             |                     |     |
| Number of ungated registers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 102 (35.54%)                    |    |                           |              |                             |              |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |              |                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |    |                           |              |                             |             |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |             |                     |     |
| Number of registers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 287                             |    |                           |              |                             |              |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |              |                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |    |                           |              |                             |             |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |             |                     |     |
| Number of Clock gating elements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 13                              |    |                           |              |                             |              |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |              |                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |    |                           |              |                             |             |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |             |                     |     |
| Number of Gated registers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 195 (73.03%)                    |    |                           |              |                             |              |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |              |                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |    |                           |              |                             |             |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |             |                     |     |
| Number of Ungated registers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 72 (26.97%)                     |    |                           |              |                             |              |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |              |                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |    |                           |              |                             |             |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |             |                     |     |
| Total number of registers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 267                             |    |                           |              |                             |              |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |              |                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |    |                           |              |                             |             |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |             |                     |     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Actual (%) Count                |    |                           |              |                             |              |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |              |                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |    |                           |              |                             |             |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |             |                     |     |
| Number of tool-inserted clock gating elements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 13 (100.00%)                    |    |                           |              |                             |              |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |              |                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |    |                           |              |                             |             |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |             |                     |     |
| Number of pre-existing clock gating elements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0 (0.00%)                       |    |                           |              |                             |              |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |              |                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |    |                           |              |                             |             |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |             |                     |     |
| Number of gated registers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 195 (73.03%)                    |    |                           |              |                             |              |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |              |                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |    |                           |              |                             |             |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |             |                     |     |
| Number of tool-inserted gated registers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 195 (73.03%)                    |    |                           |              |                             |              |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |              |                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |    |                           |              |                             |             |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |             |                     |     |
| Number of pre-existing gated registers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0 (0.00%)                       |    |                           |              |                             |              |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |              |                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |    |                           |              |                             |             |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |             |                     |     |
| Number of ungated registers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 72 (26.97%)                     |    |                           |              |                             |              |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |              |                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |    |                           |              |                             |             |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |             |                     |     |
| Number of registers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 267                             |    |                           |              |                             |              |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |              |                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |    |                           |              |                             |             |                           |     |  |                  |                                               |              |                                              |           |                           |              |                                         |              |                                        |           |                             |             |                     |     |

## D. 03\_GATE

- Run “./01\_run\_vcs\_gate” in each directory (UMC018 / N16ADFP / N16ADFP\_cg / N16ADFP\_cg\_chip)
- Do you think it is necessary to do hold time check in gate-level?

■ Yes / □ No, why: 需要驗證STA無法完全覆蓋的狀況

3. After the gate-level simulation, we can use its waveform to do the power analysis

## E. 07\_PX

1. Prime Time Power Analysis
  - a. Run “./01\_run\_ptpx” (what we use in Lab08)
  - b. Finish the table below

|               | UMC018    | N16ADFP    | N16ADFP_cg |
|---------------|-----------|------------|------------|
| Leakage Power | 8.924e-06 | 4.8251e-07 | 4.795e-07  |
| Total Power   | 0.0306    | 8.296e-04  | 3.955e-04  |

2. Design Compiler Power Analysis
  - a. Run ./02\_run\_dcpx to convert fsdb file to saif file to know the switch activity information, read DDC file to know the current circuit and generate power report
  - b. Finish the table below

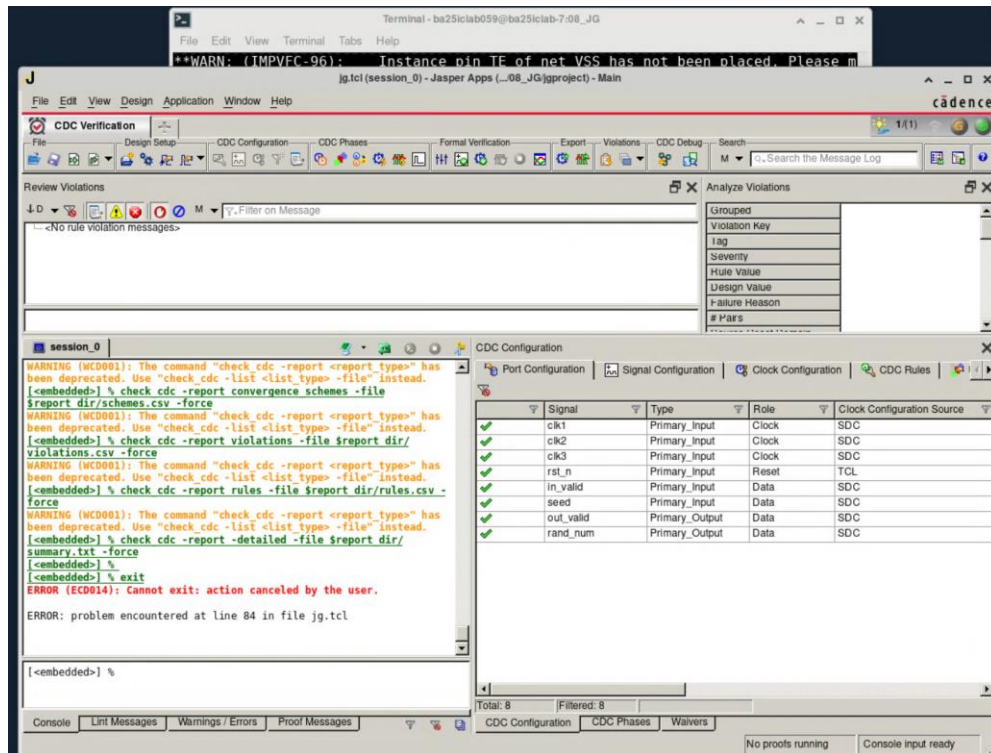
|               | UMC018   | N16ADFP  | N16ADFP_cg |
|---------------|----------|----------|------------|
| Leakage Power | 8.9307mW | 483.47nW | 480.5443nW |
| Total Power   | 30.57mW  | 0.8293mW | 0.3962mW   |

3. Notice that in ADFP, we have about 10 times faster clock speed than UMC018. Please observe the power tables above and briefly summarize the result.

N16ADFP 的時脈頻率高出 10 倍，其 Total Power (約 0.83mW) 僅為 UMC018 (30.57mW) 的 2.7%。這展現了 FinFET 製程在低電壓下的極高能效。此外，引入 Clock Gating (cg) 後，N16ADFP 的總功耗進一步下降約 52%，證明在高速設計中，動態功耗的管理也是相當重要

## F. 08\_JG

1. Run “./01\_run\_jg” (What we use in Lab07)
2. Paste your result of the CDC report



## Back-End Design Steps

### G. 05\_APR

Please answer the following questions during this lab.

#### Data Preparation

- Open **CHIP\_SHELL.v** in 05\_APR and find:
  - What is pad module name for input and output signals? PDCDG\_V、PDCDG\_H
  - What is pad module name for core power and ground? PVDD1CDGM\_V、PVDD1CDGM\_H
  - What is pad module name for I/O power and ground? PVDD2CDGM\_V、PVDD1CDGM\_H
  - Why “\_H” or “\_V” is as the suffix of pad module name?  
Poly的方向，FinFET製程的Poly不能像.18製程一樣任一方向
  - How to set the pads to avoid violation? (Hint: You can check **CHIP.io** to know the relationship between pad type and its placed location)  
上下擺Vertical的PAD 左右擺Horizontal的PAD
- Open **CHIP\_mmmc.view** in 05\_APR and find:
  - What is qrc file used in RC\_MAX? Tech/RC/N16ADFP\_QRC/worst/qrcTechFile

- b. What is qrc file used in RC\_MIN? Tech/RC/N16ADFP\_QRC/best/qrcTechFile
- c. What is PVT condition for worst case timing analysis?
  - Process corner: ss, Voltage: 0.72V, Temperature: -40° C
- d. What is PVT condition for best case timing analysis?
  - Process corner: ff, Voltage: 0.88V, Temperature: 125° C
- e. What is the sdc file used? \_CHIP.sdc
- f. How many are the setup time analysis views? 1
- g. How many are the hold time analysis views? 1

### Import Design

3. Open the **00\_import\_design.cmd** in 05\_APR/cmd and find:
  - a. The command “**set init\_design\_uniquify 1**” will make the name of every instance different so that tool can implement clock tree synthesis.
  - b. What is the design process node? 16nm
  - c. What are the lef files used?
    - Technology lef file: N16ADFP\_APR\_Innovus\_11M.10a.tlef
    - Standard cell lef file: N16ADFP\_StdCell.lef、N16ADFP\_StdIO.lef
    - Memory lef file: N16ADFP\_SRAM\_100.lef
  - d. What are the power names used for init\_pwr\_net? VDD、VDDPST

### Configuration

4. Open the **01\_config.cmd** file in 05\_APR/cmd and find:
  - a. What is the metal layer setup for topRoutingLayer? M9
  - b. What is the metal layer setup for bottomRoutingLayer? M2
  - c. What is the mode for timing analysis? onChipVariation
  - d. What are the parameters for adding extra margin to accommodate variation in timing analysis?
    - Early / late path for setup time analysis: -early 0.8 -late 1.0
    - Early / late path for hold time analysis: -early 1.0 -late 1.1
  - e. What is the purpose of “Clock Path Pessimism Removal”? 移除不可能發生的路徑
  - f. What are nondefault rules (NDR)?
    - Trunk: M8 width = 0.124, M4 spacing = 0.04
    - Leaf: M8 width = 0.062, M4 spacing = 0.08
  - g. What is route type for clock tree routing?

- ClkTrunkNDR: top preferred layer = 7, NDR rule = TrunkNDR
- ClkLeafNDR: top preferred layer = 5, NDR rule = LeafNDR
- h. What are clock max transition time?
  - (Top, Trunk, Leaf) = (0.15, 0.1, 0.1)
- i. **Early clock flow** has been activated. Typically, CTS is performed after placement, a stage where numerous clock buffer cells are inserted. Consequently, enabling early clock flow before placement becomes beneficial. This ensures that congestion measurements during the placement process consider the presence of clock trees.

### Floorplan

5. What are parameters about your floorplan?
  - a. Specify core size:
    - Aspect Ratio (H/W): 1.0
    - Core Utilization: 0.5
  - b. Specify core margins:
    - Core to Left/Right/Top/Bottom IO Boundary: 100
6. What are parameters about placement halo and routing halo for all macros
  - a. Placement halo:
    - Add/Update Halo for Top, Bottom, Left, Right: 2um
  - b. Routing halo:
    - Add/Update Halo for Halo Value: 1um
    - Add/Update Halo for Bottom Layer / Top Layer: M1/M9
7. Open **add\_IOFiller.cmd** in 05\_APR/cmd and find:
  - a. What types of pad fillers we use? PFILLER10080 、 PFILLER01008 、 PFILLER00048 、 PFILLER00001
  - b. Why we need “-fitGap true” in the command? 不用的話會DRC /LVS Violation
8. Please show the result after executing “checkFPlan” on Innovus terminal



```

Terminal - ba25iclab059@ba25iclab-7:05_APR
File Edit View Terminal Tabs Help
Area Io : IOPowerPAD2 NOT overlapped with any rows.
Area Io : IOPowerPAD3 NOT overlapped with any rows.
Area Io : COREPowerPAD0 NOT overlapped with any rows.
Area Io : COREPowerPAD1 NOT overlapped with any rows.
Area Io : COREPowerPAD2 NOT overlapped with any rows.
Area Io : COREPowerPAD3 NOT overlapped with any rows.
**ERROR: (IMPFP-7251): block/IO pad/stand Cell not on Grid error is repeated over 5 times.Trim it. If full log is need, please run checkFPlan -outFile result.txt. then check the result.txt.
Checking IO Pads out of die...
Checking constraints (guide/region/fence).....

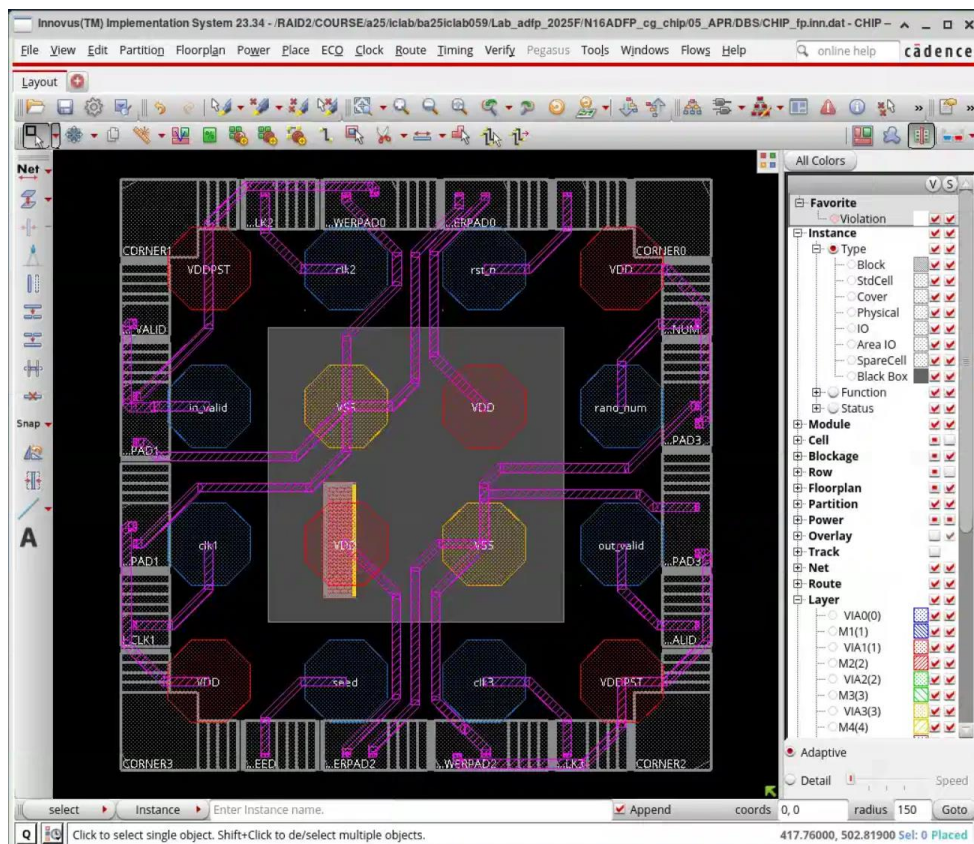
Checking Preroutes.....
No. of regular pre-routes not on tracks : 0

*** Summary of all messages that are not suppressed in this session:
Severity ID Count Summary
WARNING IMPFP-7235 1 DIE's corner: %s is NOT on %, Please u...
ERROR IMPFP-7251 1 block/IO pad/stand Cell not on Grid erro...
WARNING IMPFP-10013 2 Halo should be created around block %s a...
*** Message Summary: 3 warning(s), 1 error(s)

0
innovus 4>

```

9. What are parameters about your flip chip setting
  - a. What is the bump type we use? PAD80APB\_LF\_BU
  - b. How many bumps you use for VDD? 4
  - c. How many bumps you use for VDDPST? 2
  - d. How many bumps you use for VSS? 2
  - e. What width is AP layer used for RDL routing? 8
10. Please show your floorplan result including the bumps (Picture):



## Powerplan

11. Please list all the parameter for core power ring (You can add new sheet row)

| Nets | Metal Layer | Direction (H/V) | Width | Spacing | Interleaving | Wire group bits |
|------|-------------|-----------------|-------|---------|--------------|-----------------|
| VDD  | M10         | H               | 2     | 1       | O            | 5               |
| VSS  | M10         | H               | 2     | 1       | O            | 5               |
| VDD  | M9          | V               | 2     | 1       | O            | 5               |
| VSS  | M9          | V               | 2     | 1       | O            | 5               |
| VDD  | M8          | H               | 2     | 1       | O            | 5               |
| VSS  | M8          | H               | 2     | 1       | O            | 5               |
| VDD  | M7          | V               | 2     | 1       | O            | 5               |
| VSS  | M7          | V               | 2     | 1       | O            | 5               |
| VDD  | M6          | H               | 2     | 1       | O            | 5               |
| VSS  | M6          | H               | 2     | 1       | O            | 5               |
| VDD  | M5          | V               | 2     | 1       | O            | 5               |
| VSS  | M5          | V               | 2     | 1       | O            | 5               |
| VDD  | M4          | H               | 2     | 1       | O            | 5               |
| VSS  | M4          | H               | 2     | 1       | O            | 5               |
| VDD  | M3          | V               | 2     | 1       | O            | 5               |
| VSS  | M3          | V               | 2     | 1       | O            | 5               |

12. Please list all the parameter for block power ring (You can add new sheet row)

| Nets | Metal Layer | Direction (H/V) | Width | Spacing | Interleaving | Wire group bits |
|------|-------------|-----------------|-------|---------|--------------|-----------------|
| VDD  | M5          | V               | 0.5   | 0.5     | X            | X               |
| VSS  | M5          | V               | 0.5   | 0.5     | X            | X               |
| VDD  | M4          | H               | 0.5   | 0.5     | X            | X               |
| VSS  | M4          | H               | 0.5   | 0.5     | X            | X               |
| VDD  | M3          | V               | 0.5   | 0.5     | X            | X               |
| VSS  | M3          | V               | 0.5   | 0.5     | X            | X               |

13. Please list all the parameter for core power stripe (You can add new sheet row)

| Nets | Metal Layer | Direction (H/V) | Width | Spacing | Set-to-set | Offset | Snap |
|------|-------------|-----------------|-------|---------|------------|--------|------|
| VDD  | M10         | H               | 1.8   | 1.8     | 7.2        | 1.8    | none |
| VSS  | M10         | H               | 1.8   | 1.8     | 7.2        | 1.8    | none |
| VDD  | M9          | V               | 1.8   | 1.8     | 7.2        | 1.8    | none |



|     |    |   |       |     |      |      |           |
|-----|----|---|-------|-----|------|------|-----------|
| VSS | M9 | V | 1.8   | 1.8 | 7.2  | 1.8  | none      |
| VDD | M8 | H | 0.864 | 0   | 2.88 | 0    | half_grid |
| VSS | M8 | H | 0.864 | 0   | 2.88 | 1.44 | half_grid |
| VDD | M7 | V | 0.24  | 0   | 7.2  | 7.2  | grid      |
| VSS | M7 | V | 0.24  | 0   | 7.2  | 3.6  | grid      |
| VDD | M6 | H | 0.24  | 0   | 7.2  | 0    | grid      |
| VSS | M6 | H | 0.24  | 0   | 7.2  | 3.6  | grid      |
| VDD | M5 | V | 0.24  | 0   | 7.2  | 0    | grid      |
| VSS | M5 | V | 0.24  | 0   | 7.2  | 3.6  | grid      |

14. Please list the parameter for block power stripe (You can add new sheet row)

| Nets | Metal Layer | Direction (H/V) | Width | Spacing | Set-to-set | Offset | Snap |
|------|-------------|-----------------|-------|---------|------------|--------|------|
| VDD  | M6          | H               | 0.24  | 0       | 7.2        | 2.4    | grid |
| VSS  | M6          | H               | 0.24  | 0       | 7.2        | 1.2    | grid |
| VDD  | M5          | V               | 0.24  | 0       | 7.2        | 0      | grid |
| VSS  | M5          | V               | 0.24  | 0       | 7.2        | 3.6    | grid |

15. Please show the DRC result about your power plan (Picture)

```

Terminal - ba25iclab059@ba25iclab-7:05_APR
File Edit View Terminal Tabs Help
VERIFY DRC ..... Sub-Area : 478 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {444.416 583.296 472.192 600.744} 479 of 484
VERIFY DRC ..... Sub-Area : 479 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {472.192 583.296 499.968 600.744} 480 of 484
VERIFY DRC ..... Sub-Area : 480 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {499.968 583.296 527.744 600.744} 481 of 484
VERIFY DRC ..... Sub-Area : 481 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {527.744 583.296 555.520 600.744} 482 of 484
VERIFY DRC ..... Sub-Area : 482 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {555.520 583.296 583.296 600.744} 483 of 484
VERIFY DRC ..... Sub-Area : 483 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {583.296 583.296 601.560 600.744} 484 of 484
VERIFY DRC ..... Sub-Area : 484 complete 0 Viols.
VERIFY DRC: checking layers from AP to AP ...
VERIFY DRC ..... Using new threading
VERIFY DRC ..... Sub-Area: {0.000 0.000 601.560 600.744} 1 of 1
VERIFY DRC ..... Sub-Area : 1 complete 0 Viols.

Verification Complete : 0 Viols.

*** End Verify DRC (CPU TIME: 0:00:38.2 ELAPSED TIME: 0:00:39.0 MEM: 0.0M) ***
innovus 5>

```

16. Please show the LVS (Connectivity) result about your power plan (Picture)

```

Terminal - ba25iclab059@ba25iclab-7:05_APR
File Edit View Terminal Tabs Help
un verifyConnectivity.
**WARN: (IMPVFC-96): Instance pin TE of net VSS has not been placed. Please m
ake sure instance CORE/clk_gate_u_input_cnt_input_reg/latch is placed and rerun
verifyConnectivity.
**WARN: (IMPVFC-96): Instance pin TE of net VSS has not been placed. Please m
ake sure instance CORE/clk_gate_u_input_seed_out_reg/latch is placed and rerun v
erifyConnectivity.
Net VSS: has special routes with opens, has regular routing with opens.

Begin Summary
  2 Problem(s) (IMPVFC-200): Special Wires: Pieces of the net are not connecte
d together.
  52 Problem(s) (IMPVFC-92): Pieces of the net are not connected together.
  54 total info(s) created.
End Summary

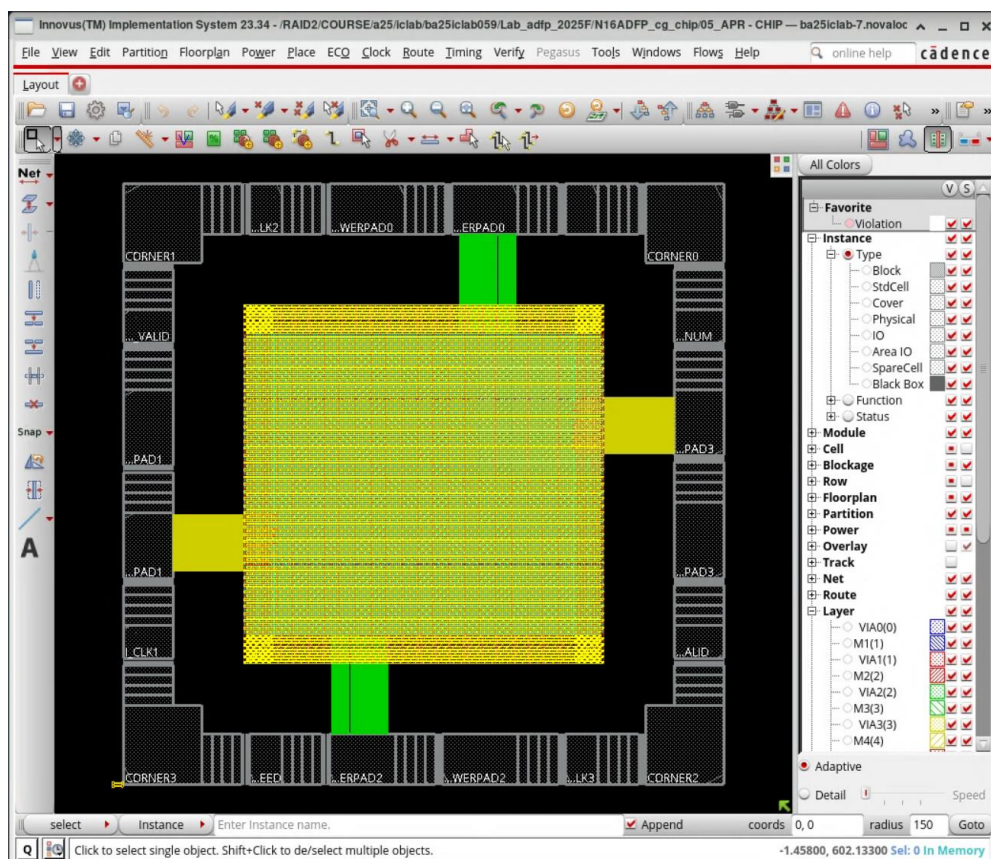
End Time: Sun Jan  4 08:29:01 2026
Time Elapsed: 0:00:01.0

***** End: VERIFY CONNECTIVITY *****
Verification Complete : 54 Viols.  0 Wrngs.
(CPU Time: 0:00:00.3 MEM: -0.820M)
Innovus 5>

```

17. How many are violations of maximum metal density? \_\_\_\_\_ 194 \_\_\_\_\_

18. Please show your power plan (Picture):



## Placement

19. Open **add\_Endcap\_WellTap.cmd** file in 05\_APR/cmd and find and find:
  - a. What is the endcap cell used at right edge? **BOUNDARY\_LEFTBWP20P90LVT**
  - b. What is the rule (or spacing) of well tap cell? \_\_\_\_\_  
 \_\_\_\_\_ **TAPCELLBWP20P90LVT** rule 40.00 \_\_\_\_\_
20. What is the placement density? \_\_\_\_\_ 0.518% \_\_\_\_\_

21. Setup time analysis in pre-CTS stage after optimization

a. What are the worst negative slack (WNS) and the total negative slack (TNS)?

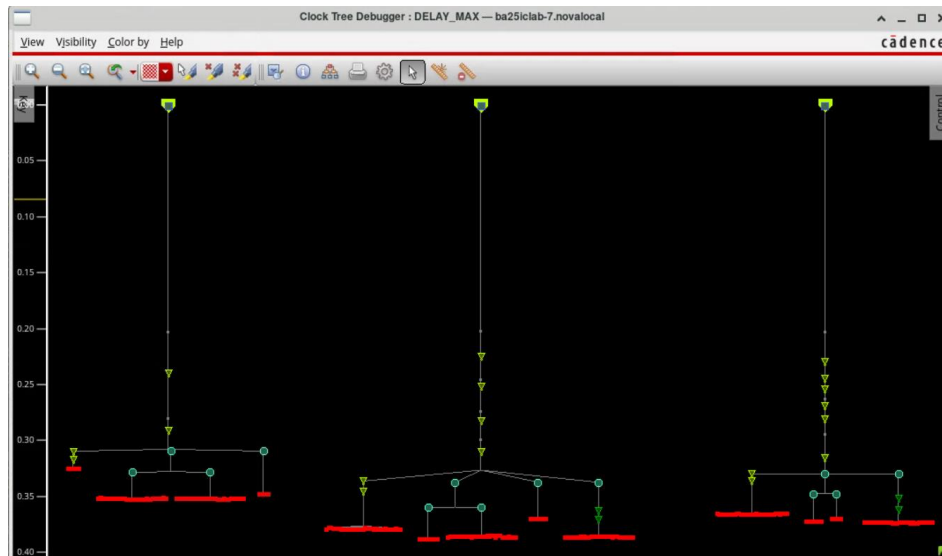
➤ WNS= -0.001, TNS= -0.001

b. How many real design rule violations (DRVs) are there?

➤ DRVs (max\_cap, max\_tran, max\_fanout) = ( 0, 0, 0 )

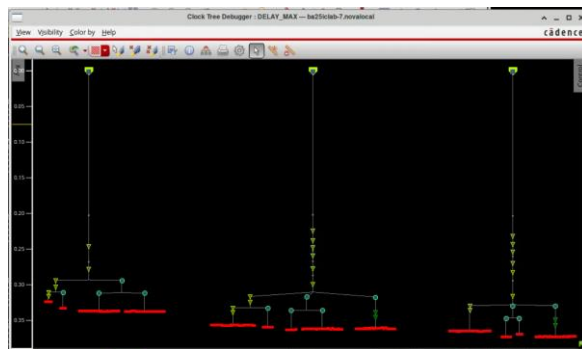
## CCOPT

22. Please show the clock tree with Clock Tree Debugger (Picture)



What is the maximum clock skew of the design after CTS?

0.388 (PostCTS 後:0.3725)



23. Setup time analysis in post-CTS stage after optimization

a. What is the worst negative slack (WNS) and the total negative slack (TNS)?

➤ WNS= -0.009, TNS= -0.009

b. How many real design rule violations (DRVs) are there?

➤ DRVs (max\_cap, max\_tran, max\_fanout) = ( 0, 0, 0 )

24. Hold time analysis in post-CTS stage after optimization

a. What is the worst negative slack (WNS) and the total negative slack (TNS)?

➤ WNS= 0.001, TNS= 0.000

b. How many real design rule violations (DRVs) are there?

➤ DRVs (max\_cap, max\_tran, max\_fanout) = ( \_\_0\_\_, \_\_0\_\_, \_\_0\_\_ )

25. Open **add\_TieHiLo.cmd** file in 05\_APR/cmd and find and find:

- What are cells for tie high / low? \_\_TIEHBWP20P90/ TIELBWP20P90\_\_
- What is max distance between each cell? \_\_100\_\_

## Nano-Route

26. Please show the LVS (Connectivity) result about your design (Picture)

```

Terminal - ba25iclab059@ba25iclab-7:05_APR
File Edit View Terminal Tabs Help

Please make sure it is assigned and rerun verifyConnectivity.
**WARN: (IMPVFC-97): IO pin seed of net seed has not been assigned. Please make
sure it is assigned and rerun verifyConnectivity.
**WARN: (IMPVFC-97): IO pin out_valid of net out_valid has not been assigned.
Please make sure it is assigned and rerun verifyConnectivity.
**WARN: (IMPVFC-97): IO pin rand_num of net rand_num has not been assigned. P
lease make sure it is assigned and rerun verifyConnectivity.
**WARN: (IMPVFC-97): IO pin VDDPST of net VDDPST has not been assigned. Pleas
e make sure it is assigned and rerun verifyConnectivity.
Net VDDPST: no routing.

Begin Summary
  1 Problem(s) (IMPVFC-98): Net has no global routing and no special routing.
  1 total info(s) created.
End Summary

End Time: Sun Jan  4 09:13:35 2026
Time Elapsed: 0:00:00.0

***** End: VERIFY CONNECTIVITY *****
Verification Complete : 1 Viols.  0 Wrngs.
(CPU Time: 0:00:00.0  MEM: 0.000M)

innovus 9>

```

27. Please show the DRC result about your design (Picture)

```

Terminal - ba25iclab059@ba25iclab-7:05_APR
File Edit View Terminal Tabs Help

VERIFY DRC ..... Sub-Area : 478 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {444.416 583.296 472.192 600.744} 479 of 484
VERIFY DRC ..... Sub-Area : 479 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {472.192 583.296 499.968 600.744} 480 of 484
VERIFY DRC ..... Sub-Area : 480 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {499.968 583.296 527.744 600.744} 481 of 484
VERIFY DRC ..... Sub-Area : 481 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {527.744 583.296 555.520 600.744} 482 of 484
VERIFY DRC ..... Sub-Area : 482 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {555.520 583.296 583.296 600.744} 483 of 484
VERIFY DRC ..... Sub-Area : 483 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {583.296 583.296 601.560 600.744} 484 of 484
VERIFY DRC ..... Sub-Area : 484 complete 0 Viols.
VERIFY DRC: checking layers from AP to AP ...
VERIFY DRC ..... Using new threading
VERIFY DRC ..... Sub-Area: {0.000 0.000 601.560 600.744} 1 of 1
VERIFY DRC ..... Sub-Area : 1 complete 0 Viols.

Verification Complete : 0 Viols.

*** End Verify DRC (CPU TIME: 0:00:38.4  ELAPSED TIME: 0:00:38.0  MEM: 256.1M)
***

innovus 9>

```

28. Setup timing analysis at post-route stage after optimization

- What are the worst negative slack (WNS) and the total negative slack (TNS)?

➤ WNS=\_\_0.009\_\_, TNS=\_\_0.000\_\_

- How many real design rule violations (DRVs) are there?

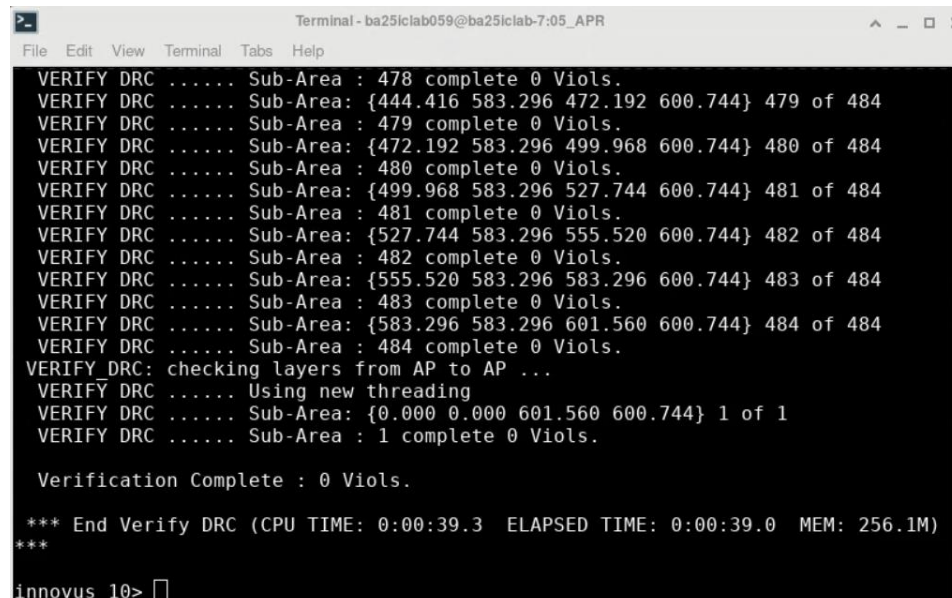
➤ DRVs (max\_cap, max\_tran, max\_fanout) = ( \_\_0\_\_, \_\_0\_\_, \_\_0\_\_ )



29. Hold timing analysis at postRoute stage after optimization
- What are the worst negative slack (WNS) and the total negative slack (TNS)?  
➤ WNS= 0.001, TNS= 0.000
  - How many real design rule violations (DRVs) are there?  
➤ DRVs (max\_cap, max\_tran, max\_fanout) = ( 0, 0, 0 )

### Finish Design

30. Open **add\_CoreFiller.cmd** in 05\_APR/cmd and find:
- What characteristic does the core filler cell “FILL32BWP20P90LVT” have?  
➤ Size / Well Tap / poly length / VT = 32 / Non-Tap/90/LVT
  - Why we need “-fitGap true” in the command? 填滿所有空隙，避免DRC
31. Please show the DRC result about your design (Picture)



```

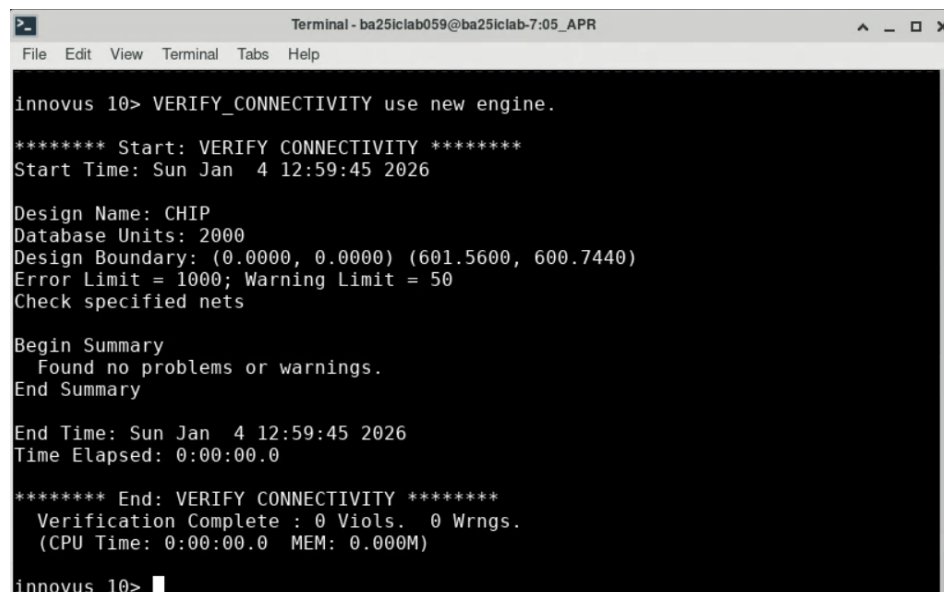
Terminal - ba25iclab059@ba25iclab-7:05_APR
File Edit View Terminal Tabs Help
VERIFY DRC ..... Sub-Area : 478 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {444.416 583.296 472.192 600.744} 479 of 484
VERIFY DRC ..... Sub-Area : 479 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {472.192 583.296 499.968 600.744} 480 of 484
VERIFY DRC ..... Sub-Area : 480 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {499.968 583.296 527.744 600.744} 481 of 484
VERIFY DRC ..... Sub-Area : 481 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {527.744 583.296 555.520 600.744} 482 of 484
VERIFY DRC ..... Sub-Area : 482 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {555.520 583.296 583.296 600.744} 483 of 484
VERIFY DRC ..... Sub-Area : 483 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {583.296 583.296 601.560 600.744} 484 of 484
VERIFY DRC ..... Sub-Area : 484 complete 0 Viols.
VERIFY DRC: checking layers from AP to AP ...
VERIFY DRC ..... Using new threading
VERIFY DRC ..... Sub-Area: {0.000 0.000 601.560 600.744} 1 of 1
VERIFY DRC ..... Sub-Area : 1 complete 0 Viols.

Verification Complete : 0 Viols.

*** End Verify DRC (CPU TIME: 0:00:39.3 ELAPSED TIME: 0:00:39.0 MEM: 256.1M)
***
innovus 10>

```

32. Please show the LVS (Connectivity) result about your design (Picture)



```

Terminal - ba25iclab059@ba25iclab-7:05_APR
File Edit View Terminal Tabs Help
innovus 10> VERIFY_CONNECTIVITY use new engine.

***** Start: VERIFY CONNECTIVITY *****
Start Time: Sun Jan 4 12:59:45 2026

Design Name: CHIP
Database Units: 2000
Design Boundary: (0.0000, 0.0000) (601.5600, 600.7440)
Error Limit = 1000; Warning Limit = 50
Check specified nets

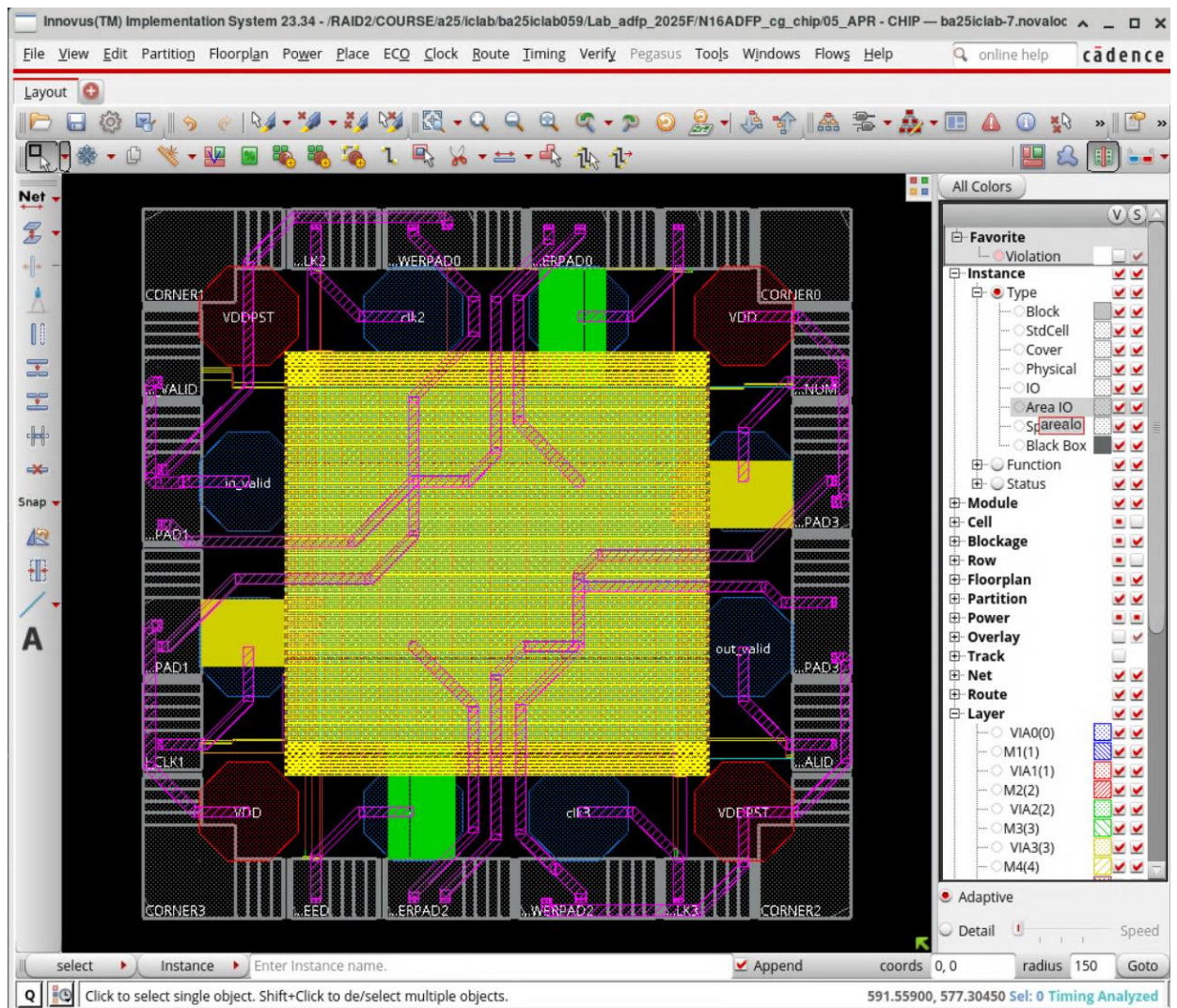
Begin Summary
Found no problems or warnings.
End Summary

End Time: Sun Jan 4 12:59:45 2026
Time Elapsed: 0:00:00.0

***** End: VERIFY CONNECTIVITY *****
Verification Complete : 0 Viols. 0 Wrngs.
(CPU Time: 0:00:00.0 MEM: 0.000M)
innovus 10>

```

33. Information after executing “verify\_antenna” on Innovus terminal
- Antenna violations: 0
34. Information after executing “CheckPlace” on Innovus terminal
- Placed = 36656 (Fixed = 5436 ), Unplaced = 0
  - Placement Density: 100%
35. Information after executing “checkFiller” on Innovus terminal
- Padded cell violations: 0
  - Gaps found: 0
36. Information after executing “VerifyMetalDensity” on Innovus terminal
- Max. Density violations: 194
37. Setup timing analysis at postRoute stage of your chip
- What is the worst negative slack (WNS) and the total negative slack (TNS)?
    - WNS= 0.001, TNS= 0
  - How many real design rule violations (DRVs) are there?
    - DRVs (max\_cap, max\_tran, max\_fanout) = ( 0, 0, 0 )
38. Hold timing analysis at postRoute stage of your chip
- What is the worst negative slack (WNS) and the total negative slack (TNS)?
    - WNS= 0.001, TNS= 0
  - How many real design rule violations (DRVs) are there?
    - DRVs (max\_cap, max\_tran, max\_fanout) = ( 0, 0, 0 )
39. Open **streamOut.cmd** file in 05\_APR/cmd and find:
- What is the map file for layout stream out?  
N16ADFP\_APR\_Innovus\_Gdsout\_11M.10a.map
  - What should .gds files be merged for layout stream out?  
N16ADFP\_StdCell.gds, N16ADFP\_StdIO.gds, N16ADFP\_BondPad.gds ,  
N16ADFP\_SRAM\_100a.gds
  - Please explain the content and the purpose of CHIP.v and CHIP\_pg.v.  
CHIP.v是 APR 完成後的 Gate-level Netlist，不包含 VDD、VSS 的。用於 gate-level simulation。  
 CHIP\_pg.v 為Power/Ground Netlist，包含了所有元件的VDD、VSS。用於 LVS驗證。
40. Please show your APR result (**Picture**)



## H. 06\_POST

1. Change the directory to **N16ADFP\_cg\_chip/06\_POST**
2. Execute the command to disable synchronizer path timing check with Prime Time

**\$ ./00\_run\_pt**

And you will get CHIP\_pt.sdf

```
Timing updates: 1 (1 implicit, 0 explicit) (0 incremental, 1 full, 0 logical)
Noise updates: 0 (0 implicit, 0 explicit) (0 incremental, 0 full)
Maximum memory usage for this session: 2988.22 MB
CPU usage for this session: 5 seconds
Elapsed time for this session: 8 seconds
Diagnostics summary: 75 errors, 24 warnings, 584 informationals
Thank you for using pt shell!
```

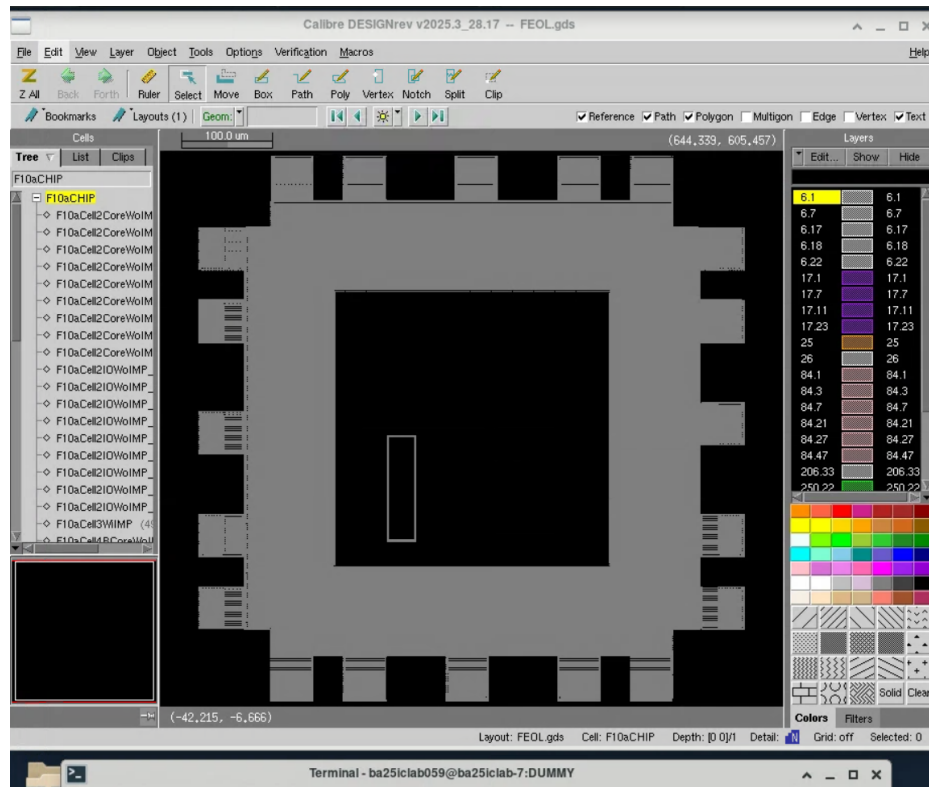
These errors are about pad cells and can be ignored in this lab

3. Execute the command to do post-layout gate-level simulation of CHIP.v

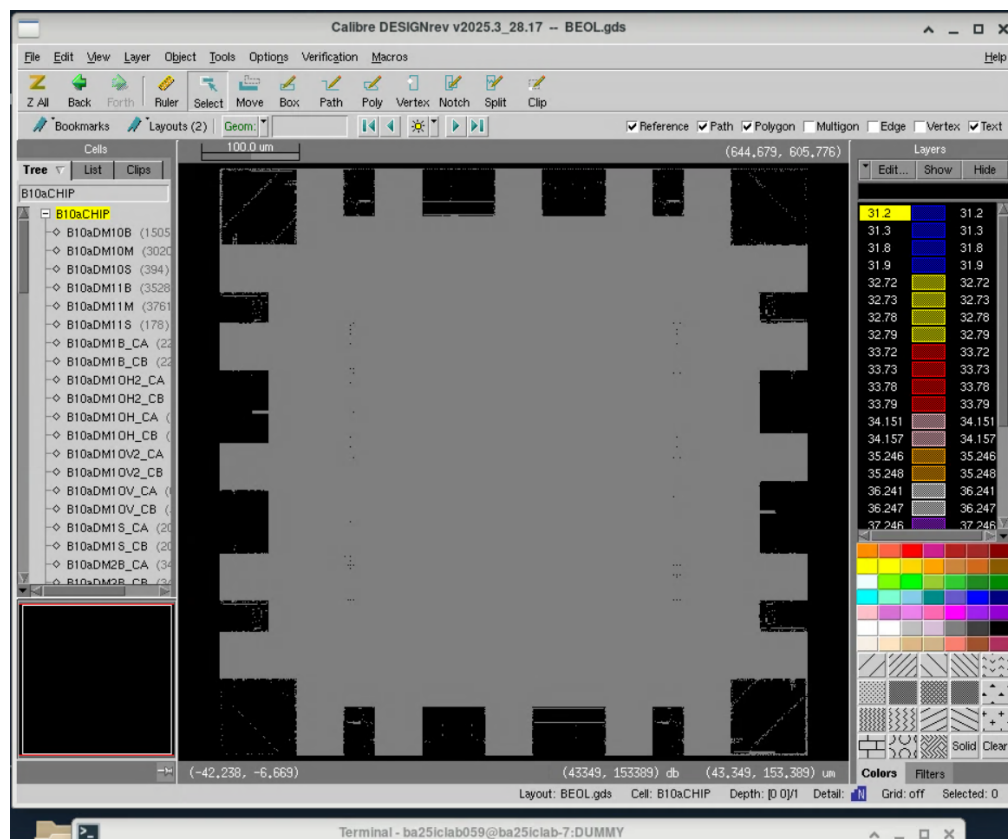
**\$ ./01\_run\_vcs\_post**

## I. 09\_RVE

1. Please show your FEOL.gds in Calibre DESIGNrev window (Picture)

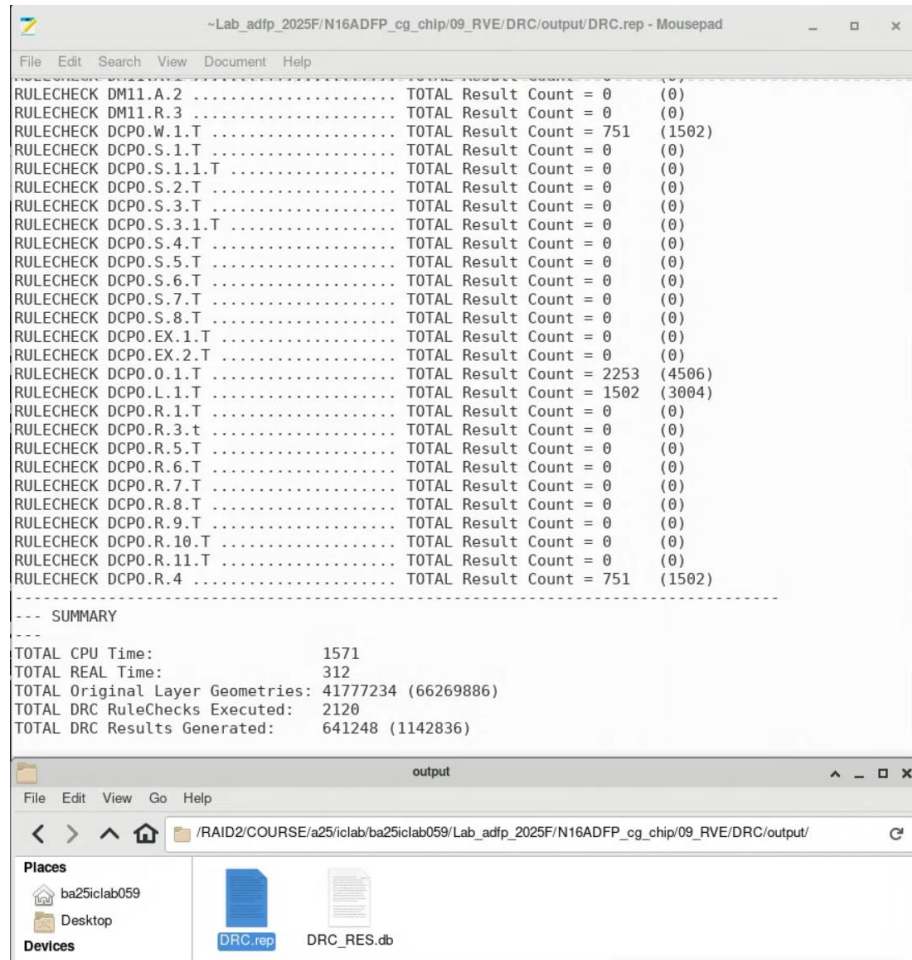


2. Please show your BEOL.gds in Calibre DESIGNrev window (Picture)





3. Please show your summary information in DRC.rep (Picture)

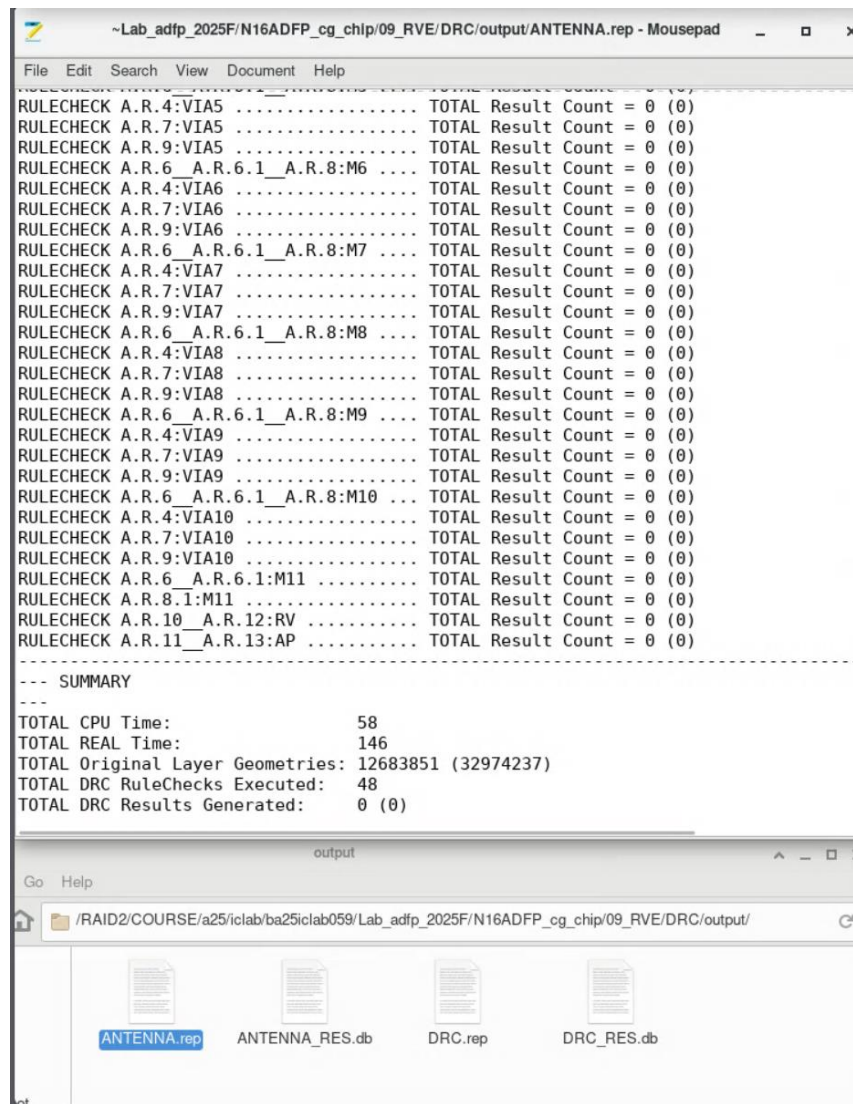


The screenshot displays two windows. The top window, titled '~Lab\_adfp\_2025F/N16ADFP\_cg\_chip/09\_RVE/DRC/output/DRC.rep - Mousepad', shows a list of DRC rule checks and their results. The bottom window, titled 'output', shows the file explorer view of the output directory, containing files 'DRC.rep' and 'DRC\_RES.db'.

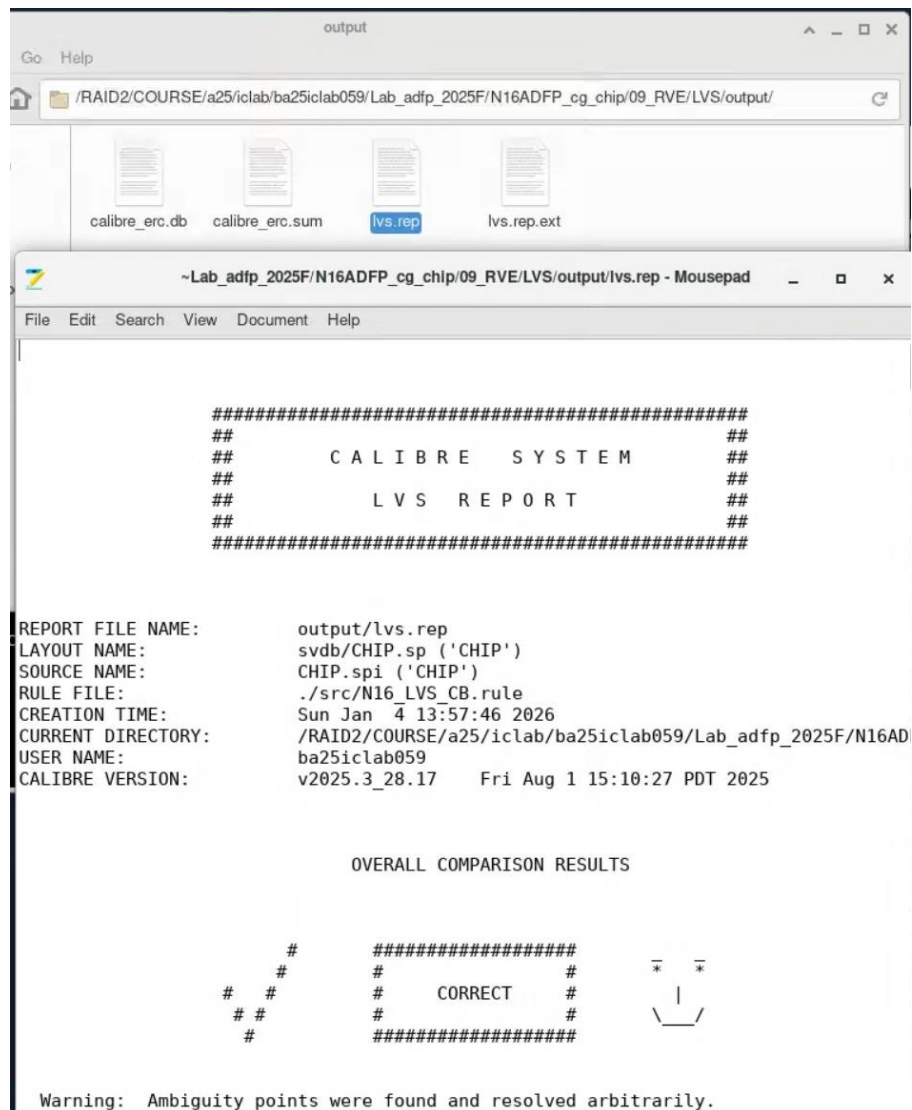
```
FILECHECK DM11.A.2 ..... TOTAL Result Count = 0 (0)
FILECHECK DM11.R.3 ..... TOTAL Result Count = 0 (0)
FILECHECK DCP0.W.1.T ..... TOTAL Result Count = 751 (1502)
FILECHECK DCP0.S.1.T ..... TOTAL Result Count = 0 (0)
FILECHECK DCP0.S.1.1.T ..... TOTAL Result Count = 0 (0)
FILECHECK DCP0.S.2.T ..... TOTAL Result Count = 0 (0)
FILECHECK DCP0.S.3.T ..... TOTAL Result Count = 0 (0)
FILECHECK DCP0.S.3.1.T ..... TOTAL Result Count = 0 (0)
FILECHECK DCP0.S.4.T ..... TOTAL Result Count = 0 (0)
FILECHECK DCP0.S.5.T ..... TOTAL Result Count = 0 (0)
FILECHECK DCP0.S.6.T ..... TOTAL Result Count = 0 (0)
FILECHECK DCP0.S.7.T ..... TOTAL Result Count = 0 (0)
FILECHECK DCP0.S.8.T ..... TOTAL Result Count = 0 (0)
FILECHECK DCP0.EX.1.T ..... TOTAL Result Count = 0 (0)
FILECHECK DCP0.EX.2.T ..... TOTAL Result Count = 0 (0)
FILECHECK DCP0.O.1.T ..... TOTAL Result Count = 2253 (4506)
FILECHECK DCP0.L.1.T ..... TOTAL Result Count = 1502 (3004)
FILECHECK DCP0.R.1.T ..... TOTAL Result Count = 0 (0)
FILECHECK DCP0.R.3.t ..... TOTAL Result Count = 0 (0)
FILECHECK DCP0.R.5.T ..... TOTAL Result Count = 0 (0)
FILECHECK DCP0.R.6.T ..... TOTAL Result Count = 0 (0)
FILECHECK DCP0.R.7.T ..... TOTAL Result Count = 0 (0)
FILECHECK DCP0.R.8.T ..... TOTAL Result Count = 0 (0)
FILECHECK DCP0.R.9.T ..... TOTAL Result Count = 0 (0)
FILECHECK DCP0.R.10.T ..... TOTAL Result Count = 0 (0)
FILECHECK DCP0.R.11.T ..... TOTAL Result Count = 0 (0)
FILECHECK DCP0.R.4 ..... TOTAL Result Count = 751 (1502)

--- SUMMARY
---
TOTAL CPU Time: 1571
TOTAL REAL Time: 312
TOTAL Original Layer Geometries: 41777234 (66269886)
TOTAL DRC RuleChecks Executed: 2120
TOTAL DRC Results Generated: 641248 (1142836)
```

4. Please show your summary information in ANTENNA.rep (Picture)



5. Open **01\_run\_v2lvs** file in 09\_RVE/LVS and find:
  - a. What Verilog files do we need? \_\_ N16ADFP\_StdCell\_pwr.v \_ 、 \_ N16ADFP\_StdIO\_pwr.v \_ 、 \_ N16ADFP\_SRAM\_100a.v \_\_\_\_\_
  - b. What spice files do we need? \_\_ N16ADFP\_StdCell\_100b.spi \_\_ 、 \_ N16ADFP\_StdIO.spi \_\_ 、 \_ N16ADFP\_SRAM\_100a.spi \_\_
6. Please show your Calibre LVS result in lvs.rep (**Picture**)



The image shows a file explorer window at the top with the path `/RAID2/COURSE/a25/iclab/ba25iclab059/Lab_adfp_2025F/N16ADFP_cg_chip/09_RVE/LVS/output/`. It contains files `calibre_erc.db`, `calibre_erc.sum`, `lvs.rep` (selected), and `lvs.rep.ext`. Below it is a terminal window titled `~Lab_adfp_2025F/N16ADFP_cg_chip/09_RVE/LVS/output/lvs.rep - Mousepad` displaying the following text:

```
#####  
##          C A L I B R E   S Y S T E M          ##  
##          L V S   R E P O R T                   ##  
##          #####  
REPORT FILE NAME:      output/lvs.rep  
LAYOUT NAME:          svdb/CHIP.sp ('CHIP')  
SOURCE NAME:          CHIP.spi ('CHIP')  
RULE FILE:            ./src/N16_LVS_CB.rule  
CREATION TIME:        Sun Jan  4 13:57:46 2026  
CURRENT DIRECTORY:    /RAID2/COURSE/a25/iclab/ba25iclab059/Lab_adfp_2025F/N16AD  
USER NAME:            ba25iclab059  
CALIBRE VERSION:      v2025.3_28.17    Fri Aug 1 15:10:27 PDT 2025  
  
OVERALL COMPARISON RESULTS  
  
#          #####  
#          #          * *  
# #        # CORRECT  #  
# #        #          #  
#          #####  
  
Warning: Ambiguity points were found and resolved arbitrarily.
```

7. Please show your layout (CHIP\_addDummy.gds) in Virtuoso (Picture)

